Lappeenrannan teknillinen yliopisto
School of Engineering Science

Software Development Skills Full Stack, Online course

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LEARNING DIARY

7.10.2026

I started the coursework by starting with setup. Afterwards I begun watching the NodeJS video. Some of the content I was already familiar from previous courses but it worked great as a reminder since it has been a while since I worked with NodeJS.

24.10.2026

I finished the first module video and learned about different kinds of modules I had not previously used. For example, I was not familiar with crypto module before and learned a lot about how I could utilize it in my own project later on. Also, learned about different functions they come with. Afterwards I started with MongoDB module. I had already used MongoDB compass before, but I had not used the MongoDB atlas. After setting it up and watching the rest of the video I learned about different CRUD operations I could do with it with the shell. Next, I'm moving on the ExpressJS module.

26.10.2026

I completed the ExpressJS module. I previously had done some small projects with ExpressJS and the video worked as a great reminder. I previously had done some middleware but not in the same extent as was done in the video. I learned that I could do some error handling through middleware which cleaned up the code a lot. Using dedicated controller files was something new to me as well. I learned that using controller files made the code a lot more readable and easier to maintain. Most of the front end js I was already familiar with also. The EJS also was something new to me and I learned some basic syntax from the video.
Overall what I learned from that video is that I should pay more attention to readability of my code than I did before. For example using those controllers and different kinds of middleware made the code easier to maintain. Next I'm moving on to the React module.

27.10.2026

I completed the React module. Again I already had some experience from it but did also learn new things from it. I had not previously used JSON database which I found to be very useful when testing the front end of my project. Also learned about the use of props in my project and understood the structure where to pass props in files a little better than I previously did. After finishing with the module I begun the MERN-stack module.

28.10.2026

I finished the MERN-stack module and learned quite a lot. I learned about routing, component structure in front end, slices and services for the front end and how to add functioning authentication in my front end which I had previously not done. I also learned about redux toolkit to allow components to access data which was new to me. I also learned to use thunks. In my coursework for the backend I also switched from commonjs to module because I was more familiar with the syntax of it. Also learned how to deploy my webpage to Heroku. However the website required to add credit card which I did not do.
I already got a idea for the project that I will be starting today as well. My idea for the project is similar to this project but in my idea, I make app to help with my studies. In this app I could store information about my assignments, for example when they are due and what courses they are part of. I begun the project by setting up all the backend files I think I need for the project so I can start working right away tomorrow.

29.10.2026

I completed the my project today. My idea for the project was to create similar app that the Goalsetter app was, but in my project the user can track their course assignments. Basic function was that user can add courses for themselves and inside those courses they could add assignments that those courses had and also due dates for those assignments. How it works was to attach those assignments to the courses by adding the course id to assignments to tie them together. I followed the examples given in the modules to complete this project. CSS was mostly the same that the modules had, but I added more to make different courses be more visible to the user and also to help display the assignments in those courses. I had few issues with assignments disappearing from the frontend when I

refreshed the page but found out the issue was that API calls kept overwriting each other. This was fixed by adding the course id's with them so that when the page loads it loads them in correct course and this kept them from overwriting each other. Project is now finished and ready to be submitted. ReadMe file is the last thing to do with the video to explain the functions of my project.